Indium

Pure indium mold cast in a laboratory

49
In

This element is named after the vivid indigo color that its atoms emit when electricity passes through it. Indium is generally found in nature combined with a mineral containing zinc or iron. Today, it is commonly used to make touchscreens. Like tin, it is also known as a metal that produces a shrieking noise when bent.

Indigo discovery

Indium salts, which are a type of indium compound, produce an indigo-blue color when heated strongly. German chemist Ferdinand Reich, the original discoverer of indium, was color-blind and needed his colleague Hieronymus Richter to see the indigo line in indium's color spectrum. When Richter claimed the element's discovery for himself, the two men fell out.

Indigo-blue flame

Extracting indium

Traces of indium can be found in several minerals, including sphalerite, quartz, and pyrite. However, the element is often collected as a by-product in the production of zinc and copper.

Sphalerite

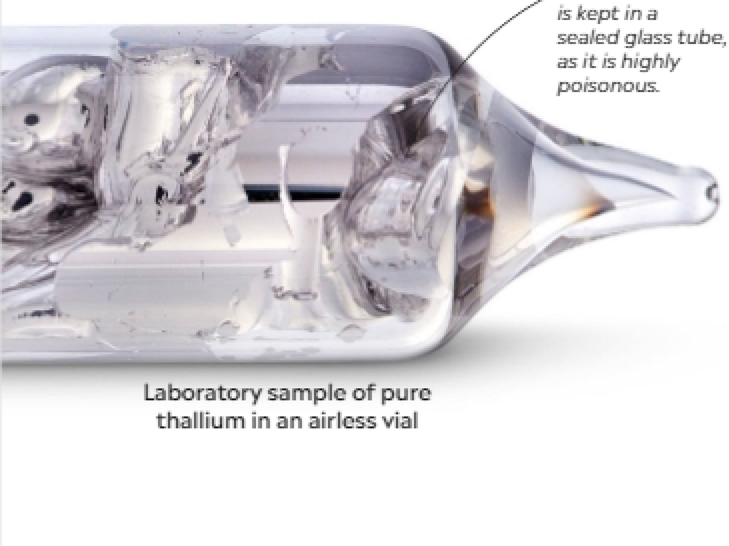
Making touchscreens

Indium is mainly used to make indium tin oxide, a compound in touchscreens and solar panels. With such a huge demand for touchscreens in modern electronics, indium's scarcity is a concern.

Thallium

81
Tl

This toxic element's name comes from *thallos*, the Greek word for a "green shoot" or "twig"—thallium emits a vibrant green color when directly inserted into a flame. Thallium was often used as a rodent poison before many accidental deaths led to a ban on its use in the US in 1972.



This soft metal is kept in a sealed glass tube, as it is highly poisonous.

Laboratory sample of pure thallium in an airless vial

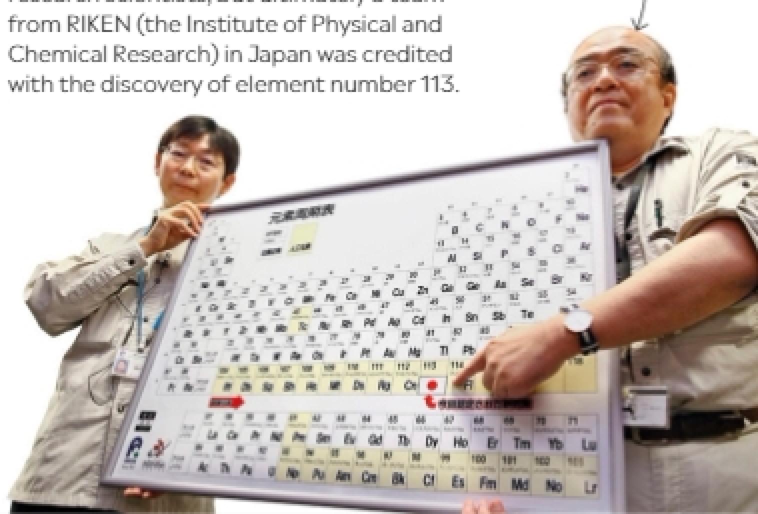
Nihonium

113
Nh

Named and placed on the periodic table in 2016, nihonium is one of the most recent additions to the table. It is named after *Nihon*, the Japanese word for "Japan," the country where the element was created. It has no known uses.

History of discovery

Nihonium was first claimed by Russian research scientists, but ultimately a team from RIKEN (the Institute of Physical and Chemical Research) in Japan was credited with the discovery of element number 113.



Professor Kōsuke Morita, who led the team at RIKEN